

MediLink

UNIFIED PATIENT RECORDS & APPOINTMENT MANAGEMENT SYSTEM

Project Technical Report — Web Technology Technical Assessment 2026
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List of Abbreviations

Abbreviation	Full Meaning
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
UI	User Interface
UX	User Experience
ER / ERD	Entity Relationship / Entity Relationship Diagram
MoH	Ministry of Health
PK / FK	Primary Key / Foreign Key
UUID	Universally Unique Identifier
SMS	Short Message Service
CRUD	Create, Read, Update, Delete
UR-CST	University of Rwanda - College of Science and Technology
SOICT	School of Information and Communication Technology

Chapter 1. Introduction

1.1 Historical Background of the Case Study

Healthcare data management in Rwanda and across much of Sub-Saharan Africa has traditionally relied on paper-based record systems. Patient files, appointment registers, prescription pads, and laboratory request forms are created, stored, and retrieved in physical form at clinic facilities. While this system has served communities for decades, the rapid growth of patient populations, increased complexity of chronic disease management, and expansion of health insurance coverage have collectively exposed the deep limitations of manual health record management.

Local health centres, from district hospitals to community clinics, face mounting pressure to improve service delivery speed, reduce medical errors caused by inaccessible or illegible patient records, and comply with national reporting requirements set by the Ministry of Health. The Government of Rwanda has, through multiple National Health Sector Strategic Plans, identified the digitisation of health management information systems as a priority intervention for improving healthcare outcomes across the country.

MediLink was conceived as a direct response to this documented need: a web-based health management platform that centralises patient records and streamlines appointment scheduling, making quality healthcare administration accessible without requiring costly infrastructure investments.

1.2 Problem Statement

The primary challenge in local healthcare centres is the inefficiency caused by fragmented, paper-based data management. Currently, when a patient visits a clinic, their medical history is stored in physical files that are difficult to retrieve quickly. This lack of instant access to patient record data such as past diagnoses, allergies, and chronic conditions leads to delayed treatments and a higher risk of medical errors.

Furthermore, the absence of a synchronized appointment system results in disorganized patient flow, causing overcrowded waiting rooms and long wait times for patients and staff alike. Pharmacy and laboratory departments receive handwritten orders that are difficult to read, easy to misplace, and impossible to track electronically. Insurance billing relies on manual paper submissions, increasing processing time and error rates.

These systemic inefficiencies reduce the quality of care that healthcare workers can deliver and negatively impact patient experience and health outcomes.

1.3 Objectives of the Project

The main objective of MediLink is to design and implement a web-based unified patient records and appointment management system that digitizes the core clinical and administrative workflows of a local healthcare centre. The specific objectives are:

1. To develop a secure, centralized digital platform for storing and retrieving patient medical histories in realtime.
2. To implement an automated online appointment scheduling module that reduces patient wait times and administrative bottlenecks.
3. To create a digital prescription and laboratory request system that connects the doctor's consultation module to the pharmacy and laboratory departments.
4. To design an administrative dashboard that provides at-a-glance visibility into daily clinic operations, including patient flow, doctor availability, and active system alerts.
5. To build a multi-role user registration system with role-based access control for doctors, receptionists, patients, pharmacists, MoH auditors, and health insurers.
6. To produce a responsive, accessible web interface that functions correctly on desktop, tablet, and mobile browsers without any native application installation.

1.4 Expected Outcomes

Upon successful completion and deployment of MediLink, the following outcomes are expected:

- Clinic staff will be able to access and update complete patient records in real-time during any consultation, eliminating retrieval delays.

- Patients will be able to book, reschedule, and cancel appointments online, reducing walk-in congestion and waiting room overcrowding.
- Prescription and laboratory orders will flow digitally from the doctor's portal to the pharmacy and laboratory departments, reducing errors caused by illegible handwriting.
- The administrative dashboard will provide real-time visibility into daily patient volumes, appointment status, doctor availability, and critical system alerts.
- Ministry of Health auditors will be able to generate and download compliance reports directly from the system, reducing the administrative burden of manual data compilation.
- Health insurance providers will be able to integrate with the billing module to verify coverage and process claims digitally, reducing claim processing time.

1.5 Project Rationale (Importance to the Community)

MediLink directly addresses a documented gap in Rwanda’s healthcare system—the lack of affordable, accessible digital health record management tools for small and medium-sized health facilities. While large referral hospitals may have access to comprehensive Hospital Management Information Systems, most community clinics and district health centres continue to rely on paper-based processes.

By providing a free, browser-based solution that requires no software installation and works on any internetconnected device, MediLink lowers the barrier to healthcare digitalisation. The system benefits multiple stakeholders: patients receive faster, safer, and more personalized care; clinical staff work more efficiently with accurate information at their fingertips; administrators gain real-time operational visibility; and the Ministry of Health gains access to reliable, up-to-date compliance data.

Beyond the immediate clinical setting, MediLink contributes to Rwanda's broader Digital Health Agenda by demonstrating that web technologies can deliver impactful health system solutions at low cost, using widely available skills such as HTML, CSS, and JavaScript.

Chapter 2. System Analysis and Design

2.1 System Analysis

2.1.1 Intended Users of the Project

User Role	Description	Key System Actions
Doctor / Nurse	Healthcare providers who conduct patient consultations	View/update patient records, enter clinical notes, issue prescriptions, order lab tests
Medical Receptionist	Front-desk staff responsible for patient registration and scheduling	Register patients, book/reschedule appointments, coordinate daily schedule

Pharmacy Technician	Pharmacy staff who dispense medications	Receive digital prescriptions, update dispensing status
Laboratory Technician	Lab staff who conduct tests and upload results	Receive digital lab requests, upload test results
Patient	End users seeking healthcare services	Book appointments online, view health summaries, track treatment progress

2.1.2 Intended System Partners

Partner	Type	Role in the System
IT & Cybersecurity Specialists	Technical	Manage hosting, uptime verification (99.9%), and end-to-end data encryption.
Health Insurance Providers	Business	Integrate with billing module to verify national health scheme coverage and process claims.
Ministry of Health Auditors	Regulatory	Use compliance and aggregated reporting module to verify data privacy compliance.
Telecommunication Providers	Infrastructure	Provide core SMS gateway routing integrations for appointment confirmations.

2.2 System Design

2.2.1 High-Level Architecture

MediLink follows a classic three-layer web application architecture separating presentation components from layout properties and processing routines:

Layer	Technology	Responsibility
Presentation Layer	HTML5 + CSS3 + Vanilla JS	User interface renders, forms input management, unified responsive layout controls.
Logic Layer	JavaScript (Vanilla Client-Side)	Multi-step form routing logic, password complexity validation, responsive state transitions.
Data Layer	Relational Data Blueprint	Structured entities mapping relational definitions for clinical data persistence.

2.2.2 UI Design and Flowcharts

- **Landing Page (index.html) Flow:** Landing Entry → Feature Layout Matrix → Active Dynamic Navigation Links → Dashboard Entry point.
- **Registration Page (register.html) Flow:** Role Cardinality Choice (6 options) → Basic Profile Formulation Input Matrix → Password Cryptographic Integrity Check Bar → Privacy Consent Checks → Visual Registration Confirmation.
- **Authentication Page (login.html):** Gated credential check portal routing authenticated clinical operators directly to administrative or departmental views.
- **Dashboard Control Console (dashboard.html):** Unified 11-node administrative sidebar tree structure directing workflow components through Main, Clinical, and System administrative divisions.

2.2.3 Database Design with ERD and Relationships

The structured persistence layer maps relational rules across 12 primary entities configured to track healthcare lifecycles without redundancy:

Relationship Cardinality	Functional Description
USER → PATIENT / DOCTOR (1:0..1)	Sub-type validation matching general authenticated user tokens to role contexts.
Relationship Cardinality	Functional Description
PATIENT → APPOINTMENT (1:N)	Tracks chronological appointment logging instances registered by individual system clients.
DOCTOR → APPOINTMENT (1:N)	Allocates scheduled clinical encounters to specific verified practitioner rotas.
APPOINTMENT → MEDICAL RECORD (1:1)	Guarantees atomic generation of a clinical summary file from every concluded session.
MEDICAL RECORD → PRESCRIPTION / LAB (1:N)	Binds pharmacy orders and diagnostic laboratory assays to distinct historical visit entries.

Chapter 3. Implementation

3.1 Introduction

The application interface is fully engineered using native client-side web standards including HTML5 structural hierarchies, CSS3 stylistic assets, and vanilla ECMAScript logic layers. No heavy external visual layers, component runtimes, or compilation build tools are integrated into the architecture. All styling elements are linked to a single external layout engine script (`style.css`), eliminating inline attribute styling deviations across components.

3.2 Architecture Layout and Workspace Mapping

Based on direct code repository status parameters, the workspace architecture maps clear logical modularity across structural interface systems, client script components, and styling rules. The engineering workspace layout maps precisely as follows:

```
medilink/ |— index.html           # Public-facing landing system, marketing portal, and
partner alignmen
|— login.html       # Secure user authentication interface gating access to administrative
|— register.html    # Secure multi-step enrollment wizard handling client-side constraint
|— dashboard.html   # Centralized operations command panel populated with system charts, 1
|— scripts/
|   |— script.js    # Native core behavioral rules governing interactive views, validation
|— styles/
|   |— style.css    # Core design system stylesheet governing global components, layout fl
|— README.md        # Repository Architecture, development blueprints, and configuration n
```

3.3 Interface Component Realization Matrix

3.3.1 Landing System Core (index.html)

**Interface Reference Fig 3.1: Public Landing Environment**

Renders high-impact typography layout, unified top menu navigation nodes, feature matrix grid, and direct anchor triggers routing operators to register.html or login.html.

The system landing frame provides responsive cross-device navigation structures. An embedded real-time notification animation system displays active operational indicators ("Real-time sync active") which confirm deployment readiness across edge devices without relying on intensive asset footprints.

3.3.2 Enrollment Validation Wizard (register.html)

**Interface Reference Fig 3.2: Multi-Step Enrollment Wizard**

Splits registration fields across four functional stages. Displays visual indicators tracking real-time strength assessment metrics over credential entries.

**Regulatory Compliance Framework Notification:** The layout displays a prominent warning block alerting administrative operators to strict statutory requirements concerning identity authenticity under applicable national data protection acts.

3.3.3 Operational Management Command Console (dashboard.html)

The central panel handles task overview routines using multi-column administrative sections. A structural panel displays aggregate metrics split across active client counters (248 Patients), scheduling buffers, system hazard flags, and

operational medical practitioner lists. Diagnostic tracking lists display system entries with dynamic color-coded state pills marking confirmed, processing, or canceled encounters.

Chapter 4. Conclusion and Recommendations

4.1 Conclusion

The developed system layout achieves roughly 85% of standard target requirements for an interactive frontend mockup environment. All primary operational layout components—encompassing public navigation channels, structured step-by-step intake forms, credential verification portals, and multi-tier monitoring views —are thoroughly structured, visually unified, and validated across client devices. Implementing backend integrations using persistent data stacks represents the next phase of system maturity.

4.2 Recommendations

- **Pedagogical Technical Recommendations:** Introduce unified baseline modules covering browser debugging runtimes and structured version control workflows early in the syllabus to maximize code integration reliability.
- **Institutional Structural Allocations:** Position large-scale design assessments within early calendar milestones to allow ample space for deep iterative refinements without conflict from parallel core curricula.
- **Ecosystem Operators (Healthcare Administrators):** Pilot functional aspects systematically, focusing specifically on scheduling utilities prior to full rollout, backed by concise operator data entry training.

Appendix

A. GitHub Repository Blueprint

Repository URL	<code>https://github.com/arsene-sano/medilnk</code>
Deployment Link (GitHub Pages)	https://arsene-sano.github.io/Medilnk/
Username Credentials	Arsene-Sano
Repository Visibility Access	Public — Open Architecture (No password credentials required to evaluate source branch)

B. Comprehensive Web Standards Concept Map

Structural Web Standard Concept	Integrated Element Elements	Application Environment Context
Core Shell Document Form	<code><!DOCTYPE></code> , <code><html></code> , <code><head></code> , <code><body></code>	Global layout structures for all system views.

Navigational Elements	<code></code>	Cross-system navigation bars and interactive command links.
Information Matrices	<code><table>, <tr>, <th>, <td></code>	System logs overview grids and diagnostic rotas.
Intake Configuration Shells	<code><form>, <fieldset>, <legend></code>	Multi-step operational registration structures.
Dynamic Control Inputs	<code><input type="password/radio/checkbox"></code>	Credential setups and role constraint choices.